

# Healthcare Cost Analysis using Excel and Power BI

## 1. Project Overview and Objective

This project focuses on cleaning, transforming, and analysing healthcare data using Excel and Power BI to generate meaningful insights through an interactive dashboard.

The main objective is to analyse hospital performance, patient discharge trends, and cost variations across hospitals and states, and to understand the gap between covered charges and actual payments.

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## 2. Data Sources

- **Source Description:** Data world and timeline not mentioned
  - **Domain:** Healthcare Analytics
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## 3. Problem Statement

- To analyse hospital discharge trends across different hospitals
  - To identify hospitals with high covered charges and cost gaps
  - To compare average covered charges with actual payments
  - To evaluate differences in healthcare cost across states
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## 4. Attribute (Column /Features) Details:

| Attribute Name           | Data Type          | Description                           |
|--------------------------|--------------------|---------------------------------------|
| DRG Definition           | Text               | Type of disease or treatment category |
| Provider Id              | Text               |                                       |
| Provider Name            | Text               | Name of the hospital                  |
| Provider City            | Text               | City where hospital is located        |
| Provider State           | Text               | State where hospital is located       |
| Total Discharges         | Numeric (Integer)  | Number of patients discharged         |
| Average Covered Charges  | Numeric (Currency) | Average charges covered               |
| Average Total Payments   | Numeric (Currency) | Total average payments                |
| Average Medical Payments | Numeric (Currency) | Payments made by insurance            |

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## 5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.

- **Power BI:** DAX calculations, visualization, and interactive dashboard creation.
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## **6. Data Pre-Processing (Power Query)**

### **Tasks Performed:**

- Removed irrelevant columns such as Summary Level, State FIPS, and Insurance demographic columns.
  - Deleted duplicate columns like Referral Region Description and Zip Code (as analysis is at city/state level).
  - Handled data formatting issues and converted columns to appropriate data types.
  - Applied filtering and sorting to organise the dataset.
  - Verified data consistency and ensured clean dataset for analysis.
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## **7. Data Handling (Power BI)**

- Used single table dataset (no relationships)
- Created simple calculated measures:

### **Measures:**

- Total Discharges
  - Average Covered Charges
  - Average Payments
  - Cost Gap = Charges – Payments
  - Patient Burden %
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## **8. Analysis and Visualizations (Power BI)**

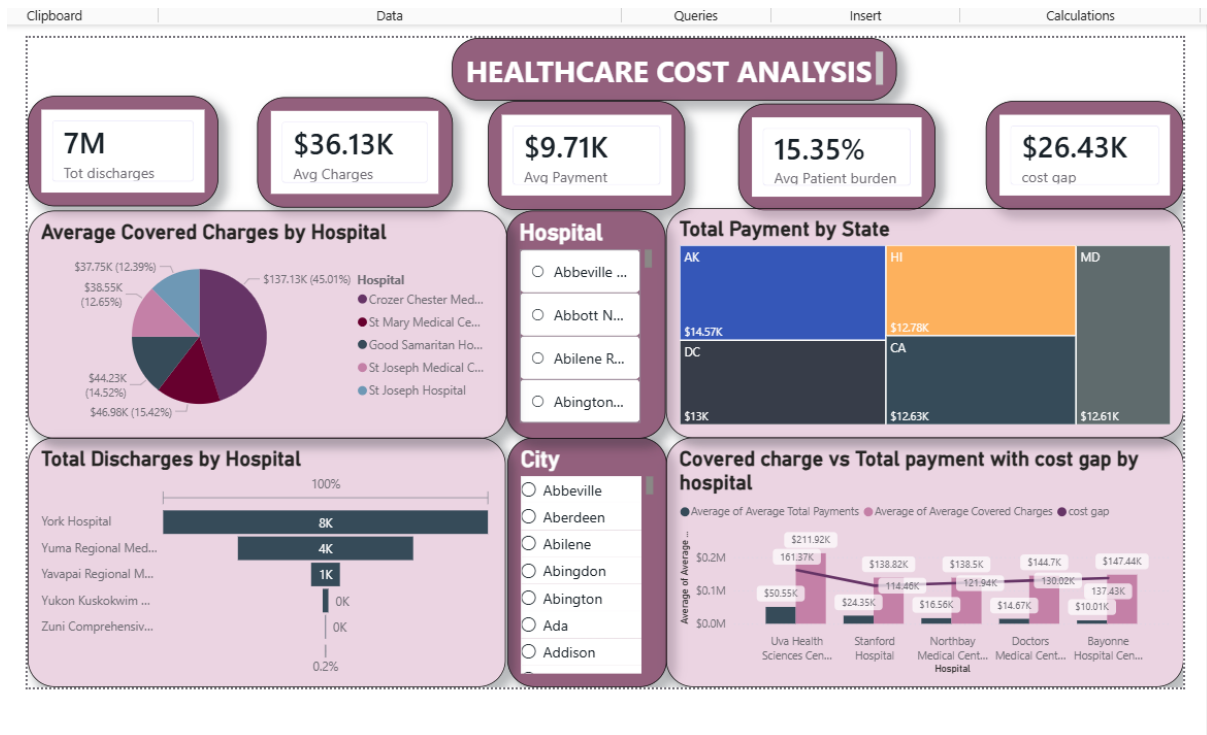
### **Dashboard Features:**

- Covered Charges by Hospital (Pie Chart)
- Total Discharges by Hospital (Bar Chart)
- Cost Gap by Hospital (Bar Chart)
- Covered Charges vs Total Payment (Combo Chart)
- Total Payment by State
- Patient Paid by State
- High Charges vs Payment Comparison

### **Features:**

- Slicers (Hospital, City)
- Interactive dashboard
- Clear titles and labels

## SCREENSHOT OF DASHBOARD



## 9. Insights & Conclusions

### Key Findings:

- There is a significant gap between covered charges and actual payments
- A few hospitals contribute disproportionately to the overall cost gap
- State-wise variations exist in total payments and patient burden
- Healthcare cost distribution is uneven across hospitals and regions
- Average hospital charges are significantly higher than actual payments
- Total discharges reached **7 million**, indicating large data volume
- Some hospitals like Crozer Chester Medical Center have very high charges
- States like Alaska and DC show higher total payments
- Large **cost gap (\$26.43K)** indicates hospitals charge much more than what is paid
- High **negotiation percentage (66%)** shows strong insurance negotiations
- Certain hospitals consistently show higher cost gaps (e.g., Uva Health Sciences Center)
- Patient burden varies by state → depends on pricing and insurance coverage
- If current trend continues, **cost gap will remain high**
- Hospitals with high charges will continue to dominate cost distribution
- Patient burden may increase in high-cost states
- Negotiation rates may stabilize but still vary across hospitals

To improve the system:

- Improve pricing transparency in hospitals
- Standardize negotiation policies

- Focus on reducing high-cost hospitals' pricing
  - Government or regulators can monitor high cost gap hospitals
  - Patients should be guided toward cost-effective hospitals
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## **Conclusion**

The analysis clearly shows a significant gap between hospital charges and actual payments. While negotiation reduces some costs, patient burden still exists.

- High-cost hospitals and states need attention
  - Cost gap is a major issue in healthcare
  - Better policies and transparency can reduce financial burden
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